

Inflow Forecasting System and Production Optimization Poso Hydropower Plant Using Bi-LSTM Seq2Seq Quantile Regression Model

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Abstract— The 515 MW Poso Hydropower Plant is crucial for Sulbagsel's grid stability, especially during peak demand (1,934 MW). However, inflow variability and reservoir constraints complicate operational planning. In 2023, an El Niño-induced drought caused a 300 MW power deficit, highlighting the need for accurate inflow forecasting. This study develops a Seq2Seq Bi-LSTM Quantile Regression model for inflow prediction and optimized hydropower scheduling. The model achieves R^2 accuracies of 90.05% (weekly) and 89% (monthly), with MAPE of 15.26% and 16.30% respectively. Deployed at PLN UP2B System Makassar since October 2024, it aids in daily, weekly, and monthly planning. As of March 2025, the system maintains an overall R^2 of 84.45% and MAPE of 29.28%. By integrating predictive modeling with adaptive scheduling, this system enhances operational efficiency and grid reliability in Sulbagsel

Keywords— Poso Hydropower Plant, Sulbagsel Grid Stability, Inflow Forecasting, Seq2Seq Bi-LSTM, Quantile Regression, Hydropower Scheduling, Operational Planning, El Niño Impact, Renewable Energy Optimization, Machine Learning in Energy Systems.

I. INTRODUCTION

Poso Hydropower Plant, the largest in the South Sulawesi System (Sulbagsel) with a 515 MW contracted capacity, plays a crucial role in grid stability, particularly during peak demand of 1934 MW. Operating under a "must-run" status, its annual reservoir management heavily depends on inflow variations and Lake Poso's elevation, influenced by dynamic hydrological conditions and evolving energy demands.

In 2023, an El Niño-induced drought significantly reduced inflow and lake levels, causing a 300 MW power deficit during peak hours and disrupting supply stability. Mitigation efforts such as Weather Modification Technology (TMC) and plant relocations proved costly and had limited long-term effectiveness. Additionally, the absence of an inflow prediction system hindered optimal operational planning during dry seasons.

To enhance reliability, an advanced inflow prediction system based on hydrological modeling is essential. Accurate forecasting would enable optimized reservoir management, maximizing production during favorable conditions while preserving water levels to sustain generation during droughts.

This study aims to develop predictive and optimization models for Poso Hydropower Plant operations. By integrating hydrological forecasting with adaptive scheduling, the system can improve efficiency, reduce power shortages, and enhance electricity supply reliability in Sulbagsel.

II. RELATED WORK

Accurate inflow forecasting is essential for optimizing hydropower operations and ensuring reservoir safety. Various studies have explored machine learning and deep learning models, particularly LSTM-based approaches, to handle the nonlinearity and nonstationarity of inflow data. Hybrid models, such as VMD-BiLSTM [2], WRF-Hydro-LSTM [3], and DEL-LSTM [4], have demonstrated superior performance over traditional methods by integrating decomposition, ensemble learning, and physically based models, achieving high predictive accuracy (e.g., NSE > 0.95, MAPE < 15%). Additionally, hybrid deep learning models like CNN-LSTM [6] have effectively captured spatial and temporal dependencies, outperforming conventional statistical methods such as ARIMA, SVM, and ANFIS. Comparative studies also highlight the limitations of standalone models, with hybrid approaches significantly improving forecasting accuracy [5]. However, existing studies have not leveraged seq2seq Bi-LSTM with quantile regression, which predicts multiple quantiles (q10, q50, q90), offering a probabilistic estimation of inflow scenarios. This novel approach enhances decision-making by accounting for uncertainty, making it particularly valuable for hydropower scheduling and reservoir management.

III. METHODOLOGY

In this study we made a Poso lake inflow forecasting system which uses the Seq2Seq Bi-LSTM Quantile Regression model, which then helps to determine the amount of Poso hydropower plant energy planning to be more efficient.

A. Study Area



Figure 1. Study Area

The Poso Hydroelectric Power Plant (PLTA Poso) is located in Poso Regency, Central Sulawesi, Indonesia. It is situated along the Poso River, which connects Lake Poso to Tomini Bay. Geographically, it lies at approximately 1°S latitude and 120°E longitude, positioned between the coastal areas of North Poso and the highlands around Lake Poso.

To support the prediction results of this study we also took 7 weather collection points or in our research we took rain data, which are in the sub-districts of tantena, panjao, mayasari, sulewana, pandiri, tindoli, and salukaia which details can be seen in Figure 1.

B. Dataset and Data Preprocessing

This study utilizes PLTA Poso data from January 1, 2012, to September 30, 2024, including rainfall (mm) from seven regions and historical inflow (m³/s) for Lake Poso. The dataset consists of 3,927 records, split into 70% training, 30% testing, and 10% of the test data used for validation. To enhance model performance, we apply RobustScaler normalization, which removes the median and scales data based on the interquartile range, ensuring a consistent range of 0–1 [7].

$$X_{Scaled} = \frac{X - Q_2}{Q_3 - Q_1} \quad (1)$$

Where X is the value to be scaled, Q_2 is median of the data, Q_1 is the first quartile (25th percentile), Q_3 is the third quartile (75th percentile).

RobustScaler enhances machine learning performance by normalizing data using the median and interquartile range (IQR), making it resistant to outliers. Unlike StandardScaler and MinMaxScaler, which rely on mean or extreme values, RobustScaler prevents outliers from skewing the scaling process. This preserves the dataset's structure, improving model accuracy, especially for algorithms sensitive to feature scaling [8].

C. Model Design

This study employs a Seq2Seq Bi-LSTM Quantile Regression model to forecast inflows under dry, normal, and wet assumptions for 7 weeks and 30 days. LSTM, an advanced RNN variant, mitigates the vanishing gradient problem by utilizing memory cells and gated mechanisms to retain long-term dependencies. These gates—forget, input, and output—regulate information flow using sigmoid activations, enabling selective data retention and efficient backpropagation. The forget gate evaluates the current input and previous hidden state to determine how much past information should be preserved [9].

$$f_t = \sigma(w_i \cdot x_t + u_f \cdot h_{t-1} + b_f) \quad (2)$$

Second, The LSTM unit determines which new information should be stored in the cell state. The input gate layer i_t , using a sigmoid activation function, identifies the values that need to be updated. Meanwhile, a $\tanh$ activation generates a vector of candidate values $\tilde{c}$ to be incorporated into the cell state.

$$i_t = \sigma(w_i \cdot x_t + u_i \cdot h_{t-1} + b_i) \quad (3)$$

$$\tilde{c} = \tanh(w_c \cdot x_t + u_c \cdot h_{t-1} + b_c) \quad (4)$$

Third, the old state c_{t-1} is updated into the new cell state c_t .

$$c_t = f_t \cdot c_{t-1} + i_t \cdot \tilde{c}_t \quad (5)$$

Finally, the output gate o_t regulates how much of the stored information is passed to the next hidden state.

$$o_t = \sigma(w_o \cdot x_t + u_o \cdot h_{t-1} + b_o) \quad (6)$$

$$h_t = o_t \cdot \tanh(c_t) \quad (7)$$

Where $(\cdot)$ is the inner products, w , and u are the weights, and b is the bias.

Where o_t represents the output gate and c_t Similarly, the backward hidden state $\bar{h}$ represents the cell state is calculated to the forward layer.

BiLSTM as shown on Figure 2 enhances the standard LSTM model by incorporating two independent LSTM networks Forward and Backward. Both layers are combined and passed to the next layer, allowing BiLSTM to capture both past and future information at each time step. In Eq. (9), an element-wise sum is used to merge the outputs of the forward and backward passes.

$$\bar{h} = o_t \cdot \tanh(c_t) \quad (8)$$

$$h_t = \bar{h}_t \oplus \vec{h}_t \quad (9)$$

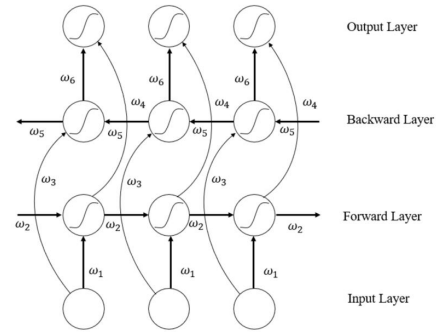


Figure 2 BiLSTM Structure

The Quantile BiLSTM model extends Bidirectional LSTM (BiLSTM) for probabilistic forecasting by predicting multiple quantiles instead of a single point estimate. This approach estimates the conditional quantile relationship between variables, capturing uncertainty in predictions. In this study, we use three quantiles 10%, 50%, and 90% representing dry, normal, and wet conditions [10][11].

$$L_q(y, \hat{y}) = \max((q - 1) \cdot (y - \hat{y}), q \cdot (y - \hat{y})) \quad (10)$$

$$L(y, \hat{Y}) = \sum_{q \in Q} L_q(y, \hat{y}_q) \quad (11)$$

Where L_q is loss for single quantile(q), L loss for all quantile, $\hat{y}$ is prediction for specific quantile.

We utilize quantile loss as shown in Eq.11 to learn the pattern of the data under the conditions described above. For more details related to the model that has been designed can be seen in Figure 3. In addition to utilizing historical data, we integrate rainfall forecast data from the OpenWeatherMap API (<https://openweathermap.org/>), which provides predictions for the next seven days. This external data source enhances the model's output by incorporating future weather conditions. To achieve this, we combine the model's predictions with future rainfall information using a simple fully connected layer, as illustrated in Figure 3. This approach ensures that the final output accounts for both historical trends and anticipated precipitation, improving the model's predictive capability.

$$Q_{out} = P \times 27.03 \quad (12)$$

Where P is the power production (MW), Q_{out} is outflow (m^3/s) and 2703 is a conversion factor.

$$V_{initial} = \frac{E_{prev} - 509.0212}{2.7747 \times 10^{-9}} \quad (13)$$

Where $V_{initial}$ is the initial lake volume (m^3), E_{prev} is the previous lake elevation (m), 509.0212 and 2.7747×10^{-9} are calibration constants.

$$V_{final} = (Q_{in} \times 86400) - (Q_{out} \times 86400) + V_{init} \quad (14)$$

Where Q_{in} prediction inflow (m^3/s) 86400 is the number of seconds in a day (to convert from m^3/s to daily volume, V_{final} is the updated lake volume (m^3).

$$E_{new} = (2.7747 \times 10^{-9} \times V_{final}) + 509.0212 \quad (15)$$

Where E_{new} is the updated lake elevation (m).

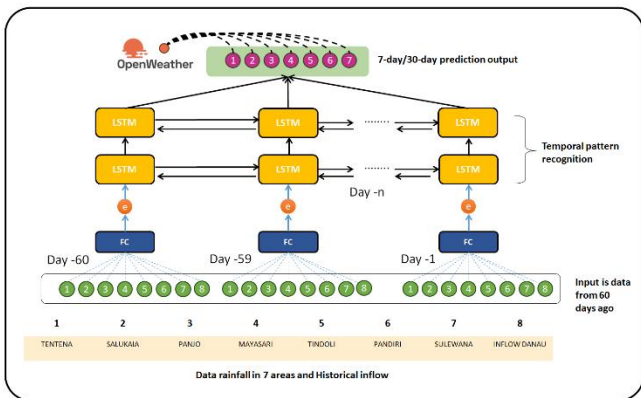


Figure 3 Model Design

The proposed model is capable of forecasting inflows at 7-day and 30-day horizons. In addition to inflow prediction, it utilizes the Poso Hydroelectric Power Plant elevation equation Eq. (15) to estimate front lake elevation, offering valuable insights for operational decision-making. By

integrating inflow forecasts with elevation dynamics, this model serves as a strategic tool for optimizing power generation at the Poso Hydroelectric Power Plant, ensuring the stability of the Sulbagsel power system while maintaining lake elevation safely above the Water Level Outage (WLO) threshold.

D. System Designed

The proposed model serves as a predictive framework capable of forecasting inflow while providing an optimized estimation of reservoir elevation and power generation to maintain system stability within the Sulbagsel grid. To ensure seamless integration and usability, we have developed an end-to-end system, as illustrated in Figure 4. This system begins by retrieving rainfall forecast data from the OpenWeather API, which is subsequently stored in a structured database using google spreadsheets.

The database not only records forecasted rainfall but also continuously updates real-time inflow and reservoir elevation data, manually inputted by the dam engineering team at the Poso Hydroelectric Power Plant. This comprehensive dataset is then fed into the predictive model, which generates inflow and elevation forecasts for the next 7 and 30 days. To enhance accessibility and usability, we have designed an interactive dashboard using Streamlit, deployed as an online platform, enabling real-time decision support for the operational planning team at PLN UP2B System Makassar.

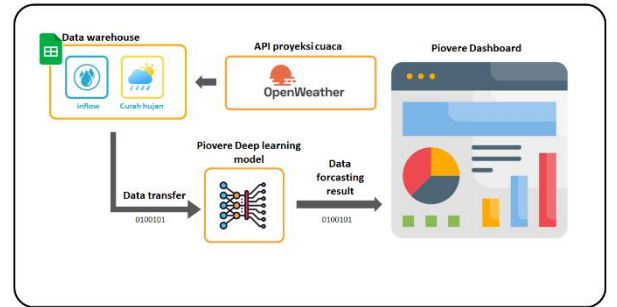


Figure 4 Model Design

IV. RESULT AND DISCUSSION

A. Model Result

The model that we have designed as shown in the image above, we design with hyper parameter settings as in Table I.

TABLE I. HYPER PARAMETER MODEL

Hyperparameter	Setting
Bi-LSTM Hidden size	128
Dropout rete	0.0049
Learning rate	0.0003
LSTM Number layer	2
Quantiles	[10%, 50%, 90%]
Optimizer type	Adam
Epoch	100
Batch size	

We trained the model using a laptop equipped with a Ryzen 7 8845HS processor and a single NVIDIA RTX 4060 (8GB) GPU. With 100 epochs, the training process was completed in approximately 1 minute and 40 seconds. The training results are presented in the Tabel II below, with the learning curve shown in Figures 7 and 8.

TABLE II. SEQ2SEQ BI-LSTM QUANTILE REGRESSION MODEL RESULT

Model Prediction Range	Quantile loss	MAPE	R^2
Week Model (7 Days)	0.18	15.26	90.05%
Month Model (30 Days)	0.189	16.30	89%

Figures 5 and 6 demonstrate consistent model performance, where both exhibit similar trends. Additionally, the R^2 values for training and validation remain relatively close, with an average difference of 6% across epochs 1 to 100.

This is also evidenced when the model is tested for wek and month models, with good results, as shown in figures 5 and 6 below.

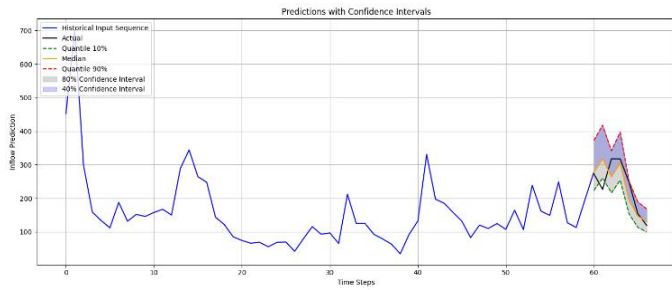


Figure 5 Model week test result

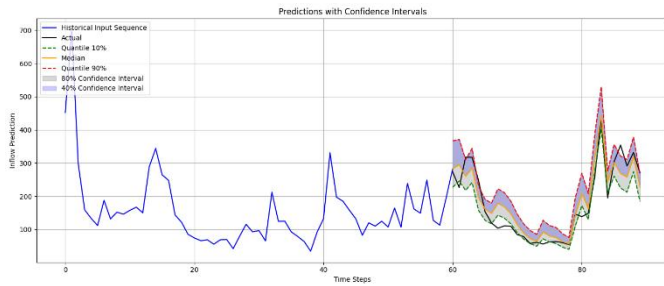


Figure 6 Model month test result

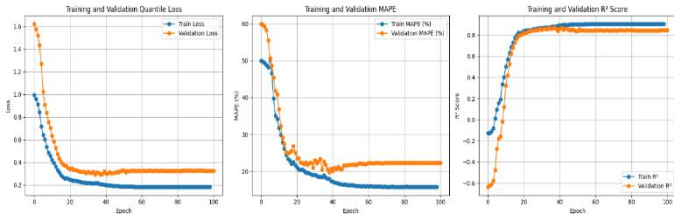


Figure 7 Learning curve Week Model (7 Days)

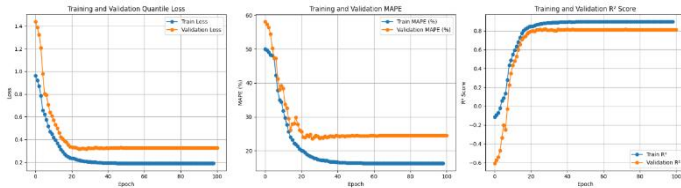


Figure 8 System Design Month Model (30 Days)

B. System Implementation result

in this study we successfully implemented the model we made with good results, with the dashboard that we designed and created using the python library, streamlit. For the results of the dashboard system can be seen in Figure 9 and 10 below. The dashboard system consists of two main sections: the settings section and the graph section. The settings section allows users to select the prediction type, either weekly or monthly, and choose the production method—manual or constrained elevation. When using the constrained elevation method, the user inputs the desired elevation value at the end of the week or month, and the system generates a recommended average production pattern to achieve the target elevation. In the manual method, users enter the production values manually and press the "Set Production" button to generate the corresponding elevation graph.

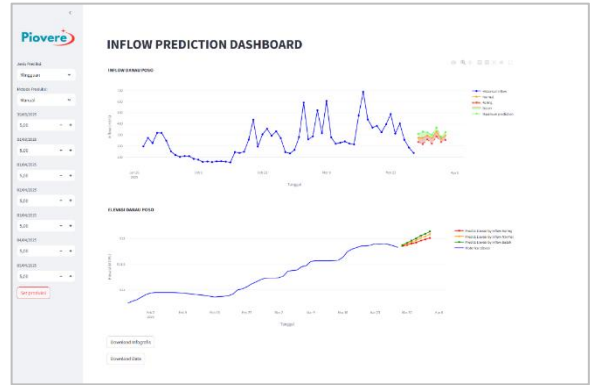


Figure 9 Dashboard Result (manual method)

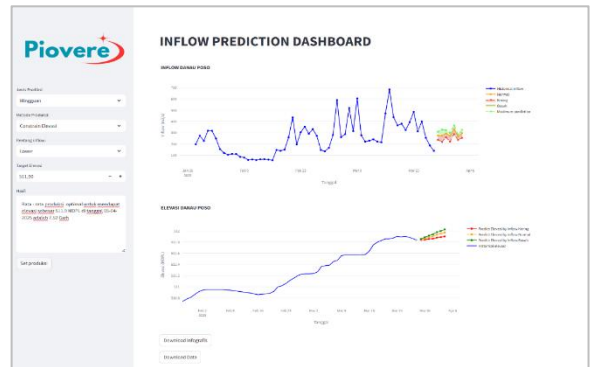


Figure 10 Dashboard Result (constrain elevation method)

This system has been successfully implemented at PLN UP2B Sistem Makassar and is utilized for the preparation of daily, weekly, and monthly operation plans by the system planning and operation evaluation team. It has been in operation since October 3, 2024, and continues to be used as of the last update on March 25, 2025, running for nearly six months. This study also includes an evaluation of the weekly prediction performance, with the results summarized in the table III below.

as of March 25, 2025, with ongoing improvements planned. The overall performance of the model shows an accuracy R^2 of 84.45% and a MAPE of 29.28%.

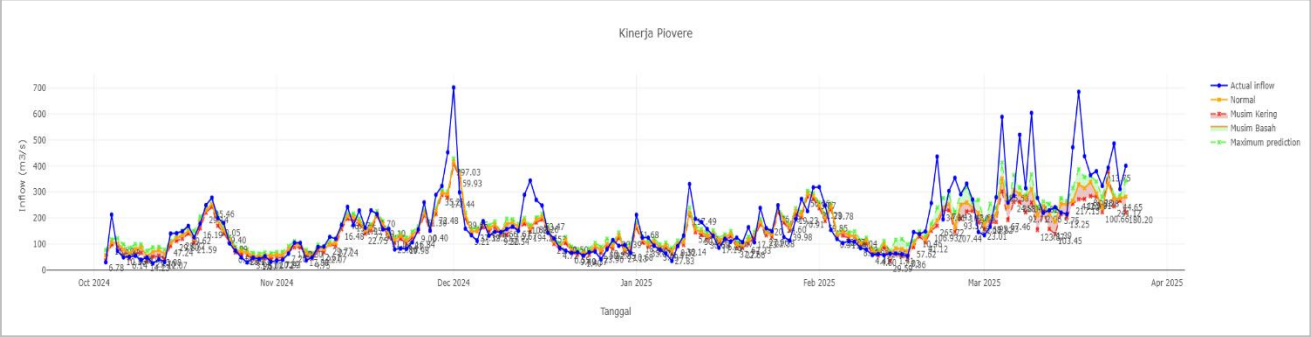


Figure 9 System evaluation

TABLE III. EVALUATION SYSTEM

Date	MAPE (%)	MAE (m ³ s)	R ² (%)
October, 2024	19.79	11.84	80.2
November, 2024	12.67	18.56	87.32
December, 2024	14.79	30.59	85.20
January, 2025	15.22	21.75	84.77
February, 2025	16.26	31.42	83.73
March, 2025	14.58	67.72	85.41
Overall	15.54	29.28	84.45

Figure 11 illustrates the actual inflow values (blue line), the wet condition range (green area), the normal inflow threshold (orange line), and the dry condition range (red area). The model demonstrates strong predictive performance for the next seven weeks, as indicated by the close alignment between actual and predicted values. The average accuracy (R^2) from October 2024 to March 25, 2025, is 84.45%, with a Mean Absolute Percentage Error (MAPE) of 15.54%, indicating a satisfactory level of accuracy.

V. CONCLUSION

This study successfully developed a predictive model and integrated it into an interactive dashboard, assisting the planning and evaluation team at PLN UP2B System Makassar in formulating daily, weekly, and monthly operational plans. Specifically, the system aids in designing the operational pattern of the Poso Hydroelectric Power Plant (PLTA Poso), one of the largest hydropower plants in the Sulbagsel system with a capacity of 515 MW. Given its crucial role in maintaining system stability, careful planning of its operation is essential, while also ensuring that the reservoir elevation remains within safe limits.

The developed model utilizes a Seq2Seq Bi-LSTM Quantile Regression approach, achieving an accuracy R^2 of 90.05% and a Mean Absolute Percentage Error (MAPE) of 15.26% for weekly predictions. For monthly predictions, the model attains an accuracy R^2 of 89% with a MAPE of 16.30%.

This system has been operational at PLN UP2B System Makassar since October 2024 and remains in continuous use

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